

Lab Exercise #3: Calculating Fare

1 Overview

Write a C program that calculates the cost of a ride based on distance, time of day, rider type, and an optional promo code.

Your program should

- Request the user to input
 - Miles traveled (a decimal number)
 - Time category (enter 1 for off-peak, 2 for peak)
 - Rider type (enter 1 for standard, 2 for student, 3 for senior)
 - Promo code (enter 0 for none, 1 for SAVE5)
- Pricing Rules
 - Base fare = \$3.00
 - Cost per mile:
 - * off-peak: \$1.80 per mile
 - * peak: \$2.40 per mile
 - Rider discount (applied to subtotal):
 - * standard: 0%
 - * student: 10%
 - * senior: 15%
 - Promo code:
 - * If promo code is SAVE5 (enter 1), subtract \$5.00 from the total **after** discounts.
 - * The final total cannot be less than \$0.00.

- Input Validation

Use `if--else` statements to validate:

- Miles must be ≥ 0
- Time category must be 1 or 2
- Rider type must be 1, 2, or 3
- Promo code must be 0 or 1

If any input is invalid, print an error message and stop.

- Output

Your program must display:

- Subtotal (before discounts)
- Discount amount
- Total after discount
- Promo applied (if any)
- Final total

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #3 - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #3 - Joe Student and Jane Doe\n");
```

3 Sample Run

Your output should match the sample run below.

```
Lab #3 -- Mark Schwitzerlett and Jacobi
February 16, 2026

Enter miles traveled: 6.5
Enter time category (1=off-peak, 2=peak): 2
Enter rider type (1=standard, 2=student, 3=senior): 2
Enter promo code (0=none, 1=SAVE5): 1

Subtotal: $18.60
Discount: $1.86
After discount: $16.74
Promo applied: -$5.00
Final total: $11.74

...Program finished with exit code 0
Press ENTER to exit console.█
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named `Lab#3_FirstName_LastName.c`

Due date: The lab is due at the end of your lab period, either Monday, February 16th or Tuesday, February 17th.

Be sure to save a copy of your .c file.